UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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QUESTION 1

a) Write the solution of 3x - 7 > -8 in interval notation and on a number line.

(3 marks)



Figure: solution set.

b) Solve the inequality |2x - 3| > 3. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Solve the equation |4x + 5| = 2.

(4 marks)

QUESTION 2

a) Apply graph-transformation techniques to sketch $y = 2(x - 1)^2 - 6$. State the vertex and the y-intercept.

(5 marks)

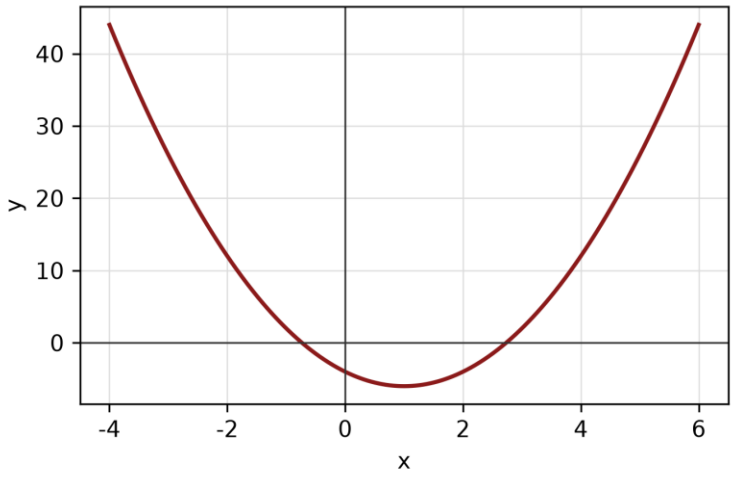


Figure: transformed graph.

b) The piecewise function is defined by
$f(x) = x - 2$, if $x < -1$;
$f(x) = -1$, if $-1 \le x \le 2$;
$f(x) = x^2 + 1$, if $x > 2$.
Sketch the graph and find f(-2), f(0) and f(3).

(5 marks)

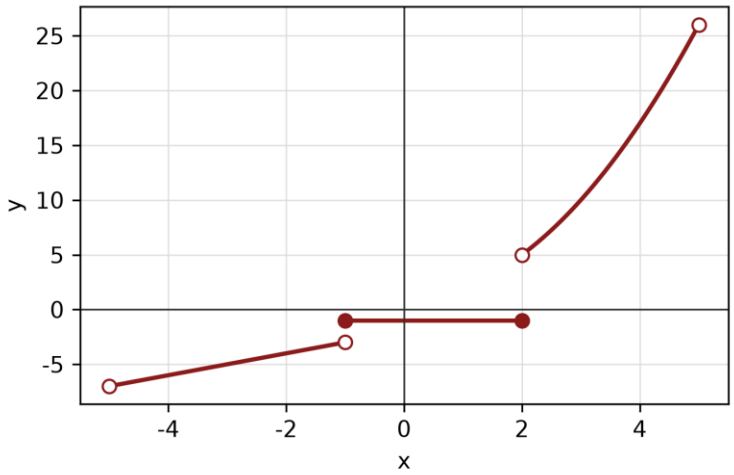


Figure: piecewise graph.

c) Find the equation of the straight line passing through (-5, 3) and (5, -4).

(4 marks)

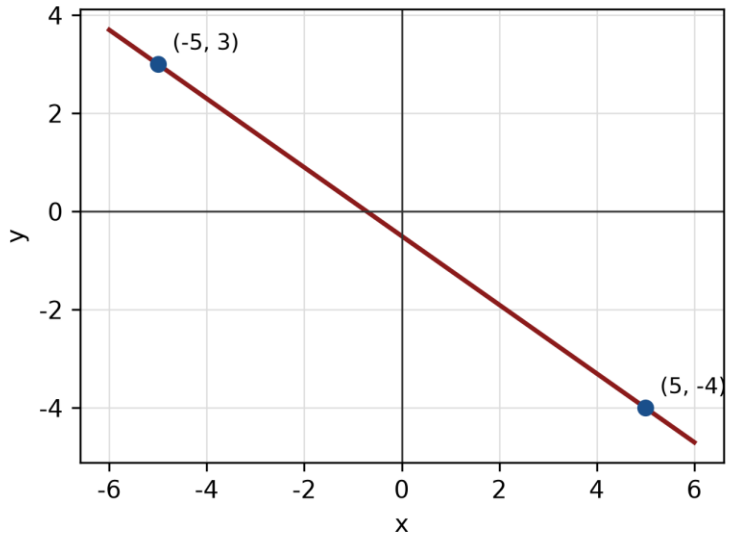


Figure: line through the two points.

QUESTION 3

a) Given $z_1 = -1 - 4i$ and $z_2 = -4 - 2i$, find $z_1 \times z_2$ in the form a + bi.

(4 marks)

b) Find the modulus and argument of $z = -5 + 12i$. Give the argument in degrees correct to one decimal place.

(5 marks)

c) Express log_5(5) + log_5(5) as a single logarithm and evaluate it.

(3 marks)

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QUESTION 4

a) Solve the simultaneous equations
$x + 2y = 1$
$5x - 5y = 20$.

(4 marks)

b) Graph the system of inequalities below and shade the feasible region. State the vertices of the region.
$x \geq 0, y \geq 0, 2x + 2y \leq 5$.

(5 marks)

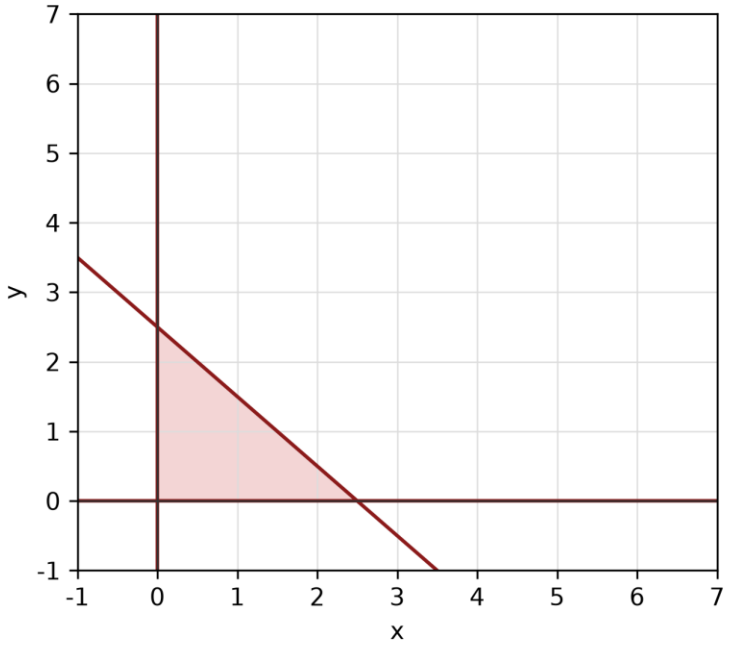


Figure: feasible region.

c) Find the standard equation of the parabola with focus (2, 1) and directrix y = 5. State its vertex.

(5 marks)

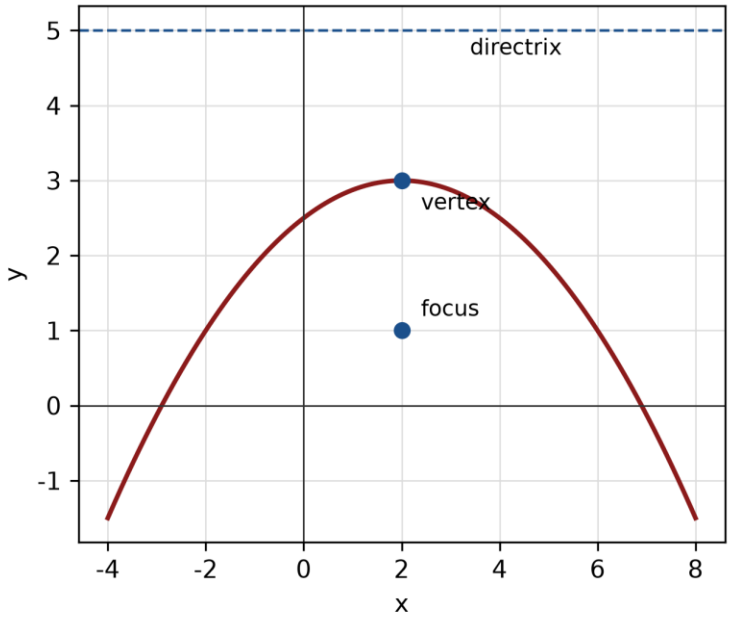


Figure: parabola with focus and directrix.

QUESTION 5

a) Prove the identity $\sin^2 x/(1 - \cos x) = 1 + \cos x$.

(5 marks)

b) Find the area of the sector of a circle with radius 8 cm and central angle 120°.

(4 marks)

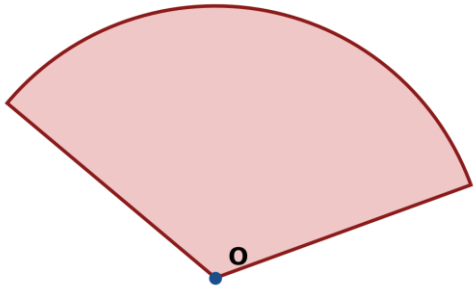


Figure: sector.

c) In triangle ABC, A = 45°, B = 60° and a = 13 cm. Use the sine rule to find b.

(4 marks)

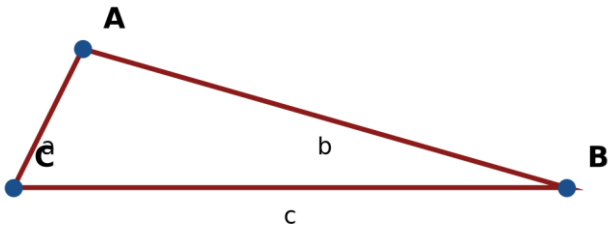


Figure: triangle ABC.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

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